Filtering numbers

INTERMEDIATE SQL

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WHERE

- `WHERE` filtering clause

WHERE

```
WHERE color = 'green'
```

WHERE with comparison operators

```
SELECT title
FROM films
WHERE release_year > 1960;
```

```
|title                 |
|----------------------|
|Judgment at Nuremberg|
|Pocketful of Miracles|
|The Hustler           |
|The Misfits           |
...
```

Comparison operators

```
SELECT title
FROM films
WHERE release_year < 1960;
```

```
|title                                          |
|-----------------------------------------------|
|Intolerance:Love's Struggle Throughout the Ages|
|Over the Hill to the Poorhouse                 |
|The Big Parade                                 |
|Metropolis                                     |
...
```

Comparison operators

```
SELECT title
FROM films
WHERE release_year <= 1960;
```

```
|title                                          |
|-----------------------------------------------|
|Intolerance:Love's Struggle Throughout the Ages|
|Over the Hill to the Poorhouse                 |
|The Big Parade                                 |
|Metropolis                                     |
...
```

Comparison operators

```
SELECT title
FROM films
WHERE release_year = 1960;
```

```
|title        |
|-------------|
|Elmer Gantry |
|Psycho       |
|The Apartment|
```

Comparison operators

```
SELECT title
FROM films
WHERE release_year <> 1960;
```

```
|title                                          |
|-----------------------------------------------|
|Intolerance:Love's Struggle Throughout the Ages|
|Over the Hill to the Poorhouse                 |
|The Big Parade                                 |
|Metropolis                                     |
...
```

Comparison operators

- `>` Greater than or after
- `<` Less than or before
- `=` Equal to
- `>=` Greater than or equal to
- `<=` Less than or equal to
- `<>` Not equal to

WHERE with strings

- Use single-quotes around strings we want to filter

```
SELECT title
FROM films
WHERE country = 'Japan';
```

```
|title            |
|-----------------|
|Seven Samurai    |
|Tora! Tora! Tora!|
|Akira            |
|Madadayo         |
|Street Fighter   |
```

Order of execution

```
-- Written code:
SELECT item
FROM coats
WHERE color = 'green'
LIMIT 5;
```

```
-- Order of execution:
SELECT item
FROM coats
WHERE color = 'green'
LIMIT 5;
```

Let's practice!

INTERMEDIATE SQL

Multiple criteria

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Multiple criteria

Multiple criteria

Multiple criteria

Multiple criteria

- `OR`, `AND`, `BETWEEN`

```
SELECT *
FROM coats
WHERE color = 'yellow' OR length = 'short';
```

```
SELECT *
FROM coats
WHERE color = 'yellow' AND length = 'short';
```

```
SELECT *
FROM coats
WHERE buttons BETWEEN 1 AND 5;
```

OR operator

- Use `OR` when you need to satisfy at least one condition

OR operator

- Correct:

```
SELECT title
FROM films
WHERE release_year = 1994
    OR release_year = 2000;
```

```
|title                      |
|---------------------------|
|3 Ninjas Kick Back         |
|A Low Down Dirty Shame     |
|Ace Ventura:Pet Detective  |
...
```

- Invalid:

```
SELECT title
FROM films
WHERE release_year = 1994 OR 2000;
```

```
argument of OR must be type boolean,
not type integer
LINE 3: WHERE release_year = 1994
OR 2000;
   ^
```

AND operator

- Use `AND` if we need to satisfy all criteria
- Correct:

```
SELECT title
FROM films
WHERE release_year > 1994
    AND release_year < 2000;
```

```
|title                        |
|-----------------------------|
|Ace Ventura:When Nature Calls|
|Apollo 13                    |
|Assassins                    |
|Babe                         |
...
```

- Invalid:

```
SELECT title
FROM films
WHERE release_year > 1994 AND < 2000;
```

```
syntax error at or near "[removed]
1994 AND < 2000;
   ^
```

AND, OR

- Filter films released in 1994 or 1995, and certified PG or R
- Enclose individual clauses in parentheses

```
SELECT title
FROM films
WHERE (release_year = 1994 OR release_year = 1995)
    AND (certification = 'PG' OR certification = 'R');
```

```
|title                 |
|----------------------|
|3 Ninjas Kick Back    |
|A Low Down Dirty Shame|
|Baby's Day Out        |
|Beverly Hills Cop III |
...
```

BETWEEN, AND

```
SELECT title
FROM films
WHERE release_year >= 1994
    AND release_year <= 2000;
```

```
SELECT title
FROM films
WHERE release_year
    BETWEEN 1994 AND 2000;
```

```
|title                    |
|-------------------------|
|3 Ninjas Kick Back       |
|A Low Down Dirty Shame   |
|Ace Ventura:Pet Detective|
|Baby's Day Out           |
...
```

```
|title                    |
|-------------------------|
|3 Ninjas Kick Back       |
|A Low Down Dirty Shame   |
|Ace Ventura:Pet Detective|
|Baby's Day Out           |
...
```

BETWEEN, AND, OR

```
SELECT title
FROM films
WHERE release_year
BETWEEN 1994 AND 2000 AND country='UK';
```

```
|title                       |
|----------------------------|
|Four Weddings and a Funeral|
|The Hudsucker Proxy         |
|Dead Man Walking            |
|GoldenEye                   |
...
```

Let's practice!

INTERMEDIATE SQL

Filtering text

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Filtering text

- `WHERE` can also filter text

```
SELECT title
FROM films
WHERE country = 'Japan';
```

```
|title             |
|------------------|
|Seven Samurai     |
|Tora! Tora! Tora!|
|Akira             |
|Madadayo          |
|Street Fighter    |
```

Filtering text

- Filter a pattern rather than specific text
- `LIKE`
- `NOT LIKE`
- `IN`

LIKE

- Used to search for a pattern in a field

% match zero, one, or many characters

```
SELECT name
FROM people
WHERE name LIKE 'Ade%';
```

```
|name          |
|--------------|
|Adel Karam    |
|Adelaide Kane|
|Aden Young    |
```

_ match a single character

```
SELECT name
FROM people
WHERE name LIKE 'Ev_';
```

```
|name          |
|--------------|
|Eve           |
```

- `Ev_ Mendes`

NOT LIKE

```
SELECT name
FROM people;
```

```
|name                |
|--------------------|
|50 Cent             |
|A. Michael Baldwin  |
|A. Raven Cruz       |
|A.J. Buckley        |
|A.J. DeLucia        |
...
```

```
SELECT name
FROM people
WHERE name NOT LIKE 'A.%';
```

```
|name           |
|---------------|
|50 Cent        |
|Aaliyah        |
|Aaron Ashmore  |
|Aaron Hann     |
...
```

Wildcard position

```
SELECT name
FROM people
WHERE name LIKE '%r';
```

```
|name           |
|---------------|
|A.J. Langer    |
|Aaron Schneider|
|Aaron Seltzer  |
|Abigail Spencer|
...
```

```
SELECT name
FROM people
WHERE name LIKE '__t%';
```

```
|name                      |
|--------------------------|
|Aitana Sánchez-Gijón      |
|Anthony 'Critic' Campos|
|Anthony Bell              |
|Anthony Burrell           |
...
```

WHERE, OR

```
SELECT title
FROM films
WHERE release_year = 1920
OR release_year = 1930
OR release_year = 1940;
```

```
|title                          |
|-------------------------------|
|Over the Hill to the Poorhouse|
|Hell's Angels                  |
|Boom Town                      |
...
```

WHERE, IN

```
SELECT title
FROM films
WHERE release_year IN (1920, 1930, 1940);
```

```
|title                          |
|-------------------------------|
|Over the Hill to the Poorhouse|
|Hell's Angels                  |
|Boom Town                      |
...
```

WHERE, IN

```
SELECT title
FROM films
WHERE country IN ('Germany', 'France');
```

```
|title         |
|--------------|
|Metropolis    |
|Pandora's Box|
|The Train     |
...
```

Let's practice!

INTERMEDIATE SQL

NULL values

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Missing values

- `COUNT(field_name)` includes only non-missing values
- `COUNT(*)` includes missing values

`null`

- Missing values:
  - Human error
  - Information not available
  - Unknown

null

```
SELECT COUNT(*) AS count_records
FROM people;
```

```
|count_records|
|-------------|
|8397         |
```

```
SELECT *
FROM people;
```

```
|id|name                |birthdate |deathdate|
|--|--------------------|----------|---------|
|1 |50 Cent             |1975-07-06|null     |
|2 |A. Michael Baldwin  |1963-04-04|null     |
|3 |A. Raven Cruz       |null      |null     |
...
```

IS NULL

```
SELECT name
FROM people
WHERE birthdate IS NULL;
```

```
|name          |
|--------------|
|A. Raven Cruz|
|A.J. DeLucia |
|Aaron Hann    |
...
```

IS NOT NULL

```
SELECT COUNT(*) AS no_birthdates
FROM people
WHERE birthdate IS NULL;
```

```
|no_birthdates|
|-------------|
|2245         |
```

```
SELECT COUNT(name) AS count_birthdates
FROM people
WHERE birthdate IS NOT NULL;
```

```
|count_birthdates|
|----------------|
|6152            |
```

COUNT() vs IS NOT NULL

```
SELECT
    COUNT(certification)
    AS count_certification
FROM films;
```

```
|count_certification|
|-------------------|
|4666               |
```

```
SELECT
    COUNT(certification)
    AS count_certification
FROM films
WHERE certification IS NOT NULL;
```

```
|count_certification|
|-------------------|
|4666               |
```

NULL put simply

- `NULL` values are missing values
- Very common
- Use `IS NULL` or `IS NOT NULL` to:
  - Identify missing values
  - Select missing values
  - Exclude missing values

Let's practice!

INTERMEDIATE SQL